

Amir Nourinia

amir.nouri.nia@gmail.com — Munich, Germany

Summary

Innovative software engineer with deep expertise in systems and graphics programming, medical imaging, and AI-driven solutions. Passionate about developing high-performance, scalable software architectures and advancing healthcare innovation.

Education

Technical University of Munich, Munich, Germany

M.Sc. in Computer Science - Biomedical Computing, 2016 – 2023

(Coursework completed; thesis incomplete)

Amirkabir University of Technology, Tehran, Iran

B.Sc. in Electrical Engineering - Bioelectronics, 2010 – 2015

Thesis: Smart MRA Injection Robot

Madani High School - National Organization of Exceptional Talents (NODET), Tabriz, Iran

High School Diploma, 2006 – 2009

Madani High School - National Organization of Exceptional Talents (NODET), Tabriz, Iran

Pre-University Diploma, 2009 – 2010

Work Experience

Holo-Light GmbH, Munich, Germany

Senior Software Engineer

07/2023 – 10/2024

- Led XR system design with a focus on low-latency streaming.
- Designed and implemented a cross-platform OpenXR library.
- Developed Android XR applications using NDK, Java/Kotlin, and MediaCodec.
- Created immersive 3D GUIs in Unity for AR/VR experiences.
- Developed a WebRTC-based real-time communication service.
- Implemented gRPC-based protocols for efficient microservice communication.
- Programmed optimized XR graphics using OpenGL ES, Vulkan, and GLSL.
- Mentored team members on C++, algorithms, and clean code practices.

One-Projects GmbH, Munich, Germany

Working Student - Software Developer

08/2022 – 03/2023

- Engaged in C++ systems programming and developed object-oriented Python libraries.
- Developed applications using Qt (C++/Qt) and ImGui (C++/OpenGL).
- Enhanced build systems with Conan and Python-based scripts.
- Implemented server load control scripts in Python.

AST (Adaptive Sensory Technology) GmbH, Munich, Germany

Working Student - Software Developer

02/2022 – 30/04/2022

- Performed C++ systems programming and graphic shader programming (GLSL/HLSL).
- Set up cross-compilation pipelines using LLVM and CMake.
- Configured remote access network infrastructure.

Fireflow GmbH, Munich, Germany

Working Student - Software Developer

04/2021 – 30/06/2021

- Developed iOS applications using SwiftUI.
- Refactored and documented legacy iOS code.
- Improved build systems and resolved user-reported issues.

SurgeVision GmbH, Munich, Germany

Working Student - Software Developer

08/2020 – 02/2021

- Worked on software engineering and algorithm development for medical image processing.
- Developed applications using Qt (C++/Qt).
- Implemented test-driven computer vision algorithms with OpenCV and GoogleTest.
- Set up Jenkins-based CI/CD pipelines.

Brainlab GmbH, Munich, Germany

Working Student - Software Developer

08/2019 – 03/2020

- Conducted research and developed algorithms for medical image processing using C++ and Python.
- Developed computer vision algorithms, performed computational modeling, and created data visualizations.

Publications

- *Design and fabrication of a nanofibrous polycaprolactone tubular nerve guide for peripheral nerve tissue engineering using a two-pole electrospinning system* (2016) [Link]
- *Orientation of directional deep brain stimulation leads on CT: Resolving the ambiguity* (2020) [Link]

Skills

Programming Languages: C++, C#, Java, Kotlin, Python, Rust, Swift

Graphics & XR: OpenGL, OpenGL ES, Vulkan, GLSL, OpenXR, Unity, Unreal Engine

Networking: WebRTC, TCP/UDP, gRPC

Software Development: Systems programming, cross-platform library design, CI/CD pipeline setup, build system optimization

Languages

- English: Proficient (C2)
- German: Intermediate (B1)
- Persian, Azerbaijani, Turkish: Native

Interests

Numerical Programming, Algorithms, Neuroscience, Computer Vision, Machine Learning, Computer Graphics, Software Engineering